Impact energy and TNT equivalents

antocciphysics | TEACHER KEY | CPA

Learning target: Convert an energy estimate without claiming a complete damage prediction.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 16; 3 course days later.*
Same work; the slower student has 1/3 of the faster student's average power.

02. Explain a worked example (worked)
Same total energy does not fix pressure, coupling, duration, material response or damage.

03. Look and explain (model)
No; both can state energy using different units.

04. Build the solution (practice)
2 tons TNT equivalent.

05. Use the idea (reason)
Effects depend on entry, altitude, angle, composition, fragmentation, energy coupling and location.

06. Choose a method (mixed) *Review: Unit 4, Day 7; 12 course days later.*
+350 J; the object gains gravitational potential energy.

07. Exit check (exit)
Both quarter.

08. STRETCH: Apply it another way (stretch)
Energy per area and coupling differ; concentrated transfer can give larger local energy density.

Teaching note: All impact values are hypothetical classroom examples. No hazardous-location or damage prediction is implied.

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01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 16; 3 course days later.*
P=Fv=150 W; W=Pt=1050 J.

02. Explain a worked example (worked)
Same total energy does not fix pressure, coupling, duration, material response or damage.

03. Represent and interpret (model)
No; both can state energy using different units.

04. Solve and justify (practice)
Speed 10^4 m/s; K 5×10^{13} J;≈11, 950 tons, or 11.95 kilotons.

05. Analyze the reasoning (reason)
Effects depend on entry, altitude, angle, composition, fragmentation, energy coupling and location.

06. Choose a method (mixed) *Review: Unit 4, Day 7; 12 course days later.*
ΔU=-350 J; gravity does +350 J work.

07. Exit check (exit)
Unchanged: total K×2, fraction×.5.

08. STRETCH: Apply it another way (stretch)
K=10^9 J; equivalent≈238 kg. Need geometry, materials, coupling, impact angle and other evidence.

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